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COURSE-END PROJECT REPORT

On

Exploratory Data Analysis and Visualization of Courier Delivery Performance

Submitted to fulfill the requirements for the Award of CIE Marks (10) for the B.E.(IT), III-SEM,
Course: **Exploratory Data Analysis & Visualization - EDAV (22ADC31N)**

By

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CERTIFICATE

This is to certify that the course-end project report entitled **"Exploratory Data Analysis and Visualization of Courier Delivery Performance"** is a bonafide record of the work carried out by **Pranav Venkat (1601-25-737-110)**, in partial fulfilment of the requirements of the course **Exploratory Data Analysis and Visualization (22ADC31N)** of the programme **B.E(IT), III-SEM**, during the academic year **2026–2027**.

The project was carried out under my guidance and supervision. To the best of my knowledge, the work reported here is original and has not been submitted to any other institution or university for the award of any degree, diploma or certificate.

Place: Hyderabad
Date: _/_/_____

Course Faculty
(Asst Prof G.Srikanth)

DECLARATION

I hereby declare that the project report entitled **"Exploratory Data Analysis and Visualization of Courier Delivery Performance"** submitted to **Dept of IT, CBIT** is a record of original work done by me under the guidance of **G.Srikanth, Asst. Professor, IT, CBIT**.

The data used was taken from the dataset supplied for the course; all external sources, tools and libraries have been acknowledged in the References. This work has not been submitted, in part or in full, for any other course, degree or diploma.

Place:

Date:

Signature of the Student
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1. Abstract

Timely and reliable delivery is a central measure of service quality in courier and logistics operations. Understanding which conditions and service choices are associated with delays helps operators plan routes, set realistic delivery promises and manage customer satisfaction. This project uses the Delivery Logistics dataset, containing 25,000 courier delivery records and 15 attributes, to examine delivery time, delay behaviour, customer rating and delivery cost, and how these relate to weather, delivery mode, delivery partner, vehicle type, package type, region, distance and package weight.

The project follows an end-to-end data analysis workflow in Python, covering data quality assessment, data cleaning, descriptive statistics, group aggregation, chi-square testing, correlation analysis and visualization. Two data quality problems were identified and handled: the delivery-time and expected-time columns were stored as timestamps and were decoded back into numeric hours, and the delivery identifier column was found to be unusable as a key. NumPy and Pandas are used for numerical computation and data manipulation, SciPy for statistical tests, and Matplotlib and Seaborn for graphical analysis.

The mean delivery time is 6.25 hours (median 6.0 hours); 26.68% of deliveries carry a delayed flag and 5.31% failed. Delay rates rise sharply in adverse weather, from 16.0% in cold weather to 41.5% in stormy weather. Delivery mode shows the strongest association with delay (Cramér's V = 0.68): 73.8% of express deliveries are delayed against 0.0% of standard deliveries, even though actual delivery time is almost identical across modes (6.23–6.28 hours). Distance is very strongly related to delivery cost ($r = 0.99$) and strongly related to delivery time ($r = 0.69$), whereas package weight is unrelated to delivery time. Differences between delivery partners, vehicle types, package types and regions are small and not statistically significant at the 5% level. Overall, the project shows how exploratory data analysis and visualization can turn raw logistics records into interpretable evidence about delivery performance.

Keywords: Courier Delivery Performance, Exploratory Data Analysis, Data Visualization, Python, Delivery Delay, Correlation Analysis.

2. Introduction

The growth of e-commerce has made courier and last-mile logistics a critical part of everyday commerce. Customers expect parcels, documents, groceries and medicines to arrive within a promised window, and logistics providers are judged on how consistently they keep that promise. Delivery time, delay rate, failed deliveries, customer rating and cost are therefore the key indicators used to judge the performance of a delivery operation.

Delivery performance varies considerably. Some deliveries arrive well within the promised time while others are late or fail altogether. This variation may be associated with several factors, including weather conditions, the delivery mode purchased by the customer, the delivery partner and vehicle used, the type and weight of the package, the region served and the distance travelled. Examining these characteristics helps reveal where delays concentrate and how much of the variation each factor can explain.

This project uses the Delivery Logistics dataset, which contains 25,000 delivery records and 15 original attributes. The dataset provides the delivery partner, package type, vehicle type, delivery mode, region, weather condition, distance, package weight, actual and expected delivery time, delay indicator, delivery status, customer rating and delivery cost. These fields support comparisons across categories and statistical examination of the relationships between distance, weight, time, cost and rating.

The project follows a systematic exploratory data analysis workflow using Python. NumPy is used for numerical computation, while Pandas supports data manipulation, data cleaning, descriptive statistics and aggregation. Matplotlib and Seaborn are used to visualize distributions, compare categories and explore relationships between variables, and SciPy is used for chi-square tests of association. The analysis is organised around the five course outcomes of the EDAV course.

The primary aim is to transform raw delivery records into meaningful and interpretable findings through statistical analysis and visualization. The analysis focuses on identifying patterns rather than establishing causal relationships. By examining delivery performance from several perspectives, the project demonstrates the practical application of exploratory data analysis techniques to a logistics problem.

3. Problem Statement and Objectives

Problem statement:

To analyze and visualize the delivery performance of a courier network by examining delivery time, delay and failure rates, customer ratings and delivery cost in terms of weather, delivery mode, delivery partner, vehicle type, package type, region, distance and package weight, and to identify the factors associated with better or worse delivery outcomes.

3.1 Objectives

1. Examine the dataset and evaluate its quality, completeness and consistency.
2. Clean and prepare the data for analysis: standardise text fields, recover numeric delivery times, handle the unusable identifier column and derive analytical features.
3. Perform Exploratory Data Analysis (EDA) on delivery time, delivery status, delay behaviour, customer rating and delivery cost.
4. Compare delivery performance across weather conditions, delivery modes, delivery partners, vehicle types, package types and regions.

5. Carry out statistical and correlation analysis: test the association of each factor with the delay flag and measure the relationships between distance, weight, time, cost and rating.
6. Visualize the results using Matplotlib and Seaborn through histograms, bar plots, scatter plots, line plots and heatmaps.
7. Present evidence-based findings, recommendations and conclusions on delivery performance, mindful of the limitations of the analysis.

3.2 Research Questions

- How are delivery times distributed, and what share of deliveries are delayed or fail?
- Which weather conditions are associated with more delays and longer delivery times?
- How do delay rates differ across delivery modes, and what explains the differences?
- Do delivery partners, vehicle types, package types or regions differ meaningfully in performance?
- How do distance and package weight relate to delivery time and delivery cost?
- How are customer ratings related to delivery outcomes?

4. Dataset Description

The analysis uses the Delivery Logistics dataset (Delivery_Logistics.csv) supplied for the course. Each row represents one courier delivery, and the dataset contains 25,000 records and 15 columns. It covers nine delivery partners, nine package types, six vehicle types, four delivery modes, five regions and six weather conditions. No calendar date or geographic coordinate is recorded for a delivery.

The principal outcome variables are the delivery status (delivered, delayed or failed), the delayed indicator, the actual delivery time in hours, the customer rating and the delivery cost. The expected delivery time represents the promised window for each delivery. Explanatory variables describe the partner, vehicle, package, mode, region, weather, distance and weight.

Before analysis, text fields were standardised, the two time columns were decoded into numeric hours, and the identifier column was examined and set aside (Section 6). The cleaned dataset was then used for descriptive analysis, category-wise comparison, statistical testing, correlation analysis and visualization.

Table 1: Columns and attributes of the dataset (as read from the CSV file)

S. No.	Column Name	Data Type	Non-Null Count	Unique Values
1	delivery_id	float64	25,000	24,502
2	delivery_partner	str	25,000	9
3	package_type	str	25,000	9
4	vehicle_type	str	25,000	6
5	delivery_mode	str	25,000	4
6	region	str	25,000	5
7	weather_condition	str	25,000	6
8	distance_km	float64	25,000	2,935
9	package_weight_kg	float64	25,000	4,853
10	delivery_time_hours	str	25,000	20

S. No.	Column Name	Data Type	Non-Null Count	Unique Values
11	expected_time_hours	str	25,000	9
12	delayed	str	25,000	2
13	delivery_status	str	25,000	3
14	delivery_rating	int64	25,000	5
15	delivery_cost	float64	25,000	22,670

Table 2: Description of the attributes

Column	Description
delivery_id	Numeric identifier (not a valid key; see Section 6.4)
delivery_partner	Courier company that handled the delivery
package_type	Category of the package (e.g. electronics, pharmacy, furniture)
vehicle_type	Vehicle used: bike, EV bike, scooter, van, EV van or truck
delivery_mode	Service level: express, same day, two day or standard
region	Region served: north, south, east, west or central
weather_condition	Weather during delivery: clear, cold, hot, foggy, rainy or stormy
distance_km	Delivery distance in kilometres
package_weight_kg	Package weight in kilograms
delivery_time_hours	Actual delivery time in hours (stored as a timestamp in the raw file)
expected_time_hours	Promised delivery time in hours (stored as a timestamp in the raw file)
delayed	Delay indicator (yes/no)
delivery_status	Outcome: delivered, delayed or failed
delivery_rating	Customer rating from 1 to 5
delivery_cost	Cost of the delivery

5. Methodology and Tools

The project follows a systematic data analysis workflow to study delivery performance. The methodology consists of data loading and inspection, data quality assessment, data cleaning and preparation, exploratory data analysis (EDA), statistical analysis, correlation analysis and visualization. Python libraries implement each step so that the analysis is consistent and reproducible.

The workflow starts by loading the CSV file into a Pandas DataFrame. The dataset is inspected for its structure, column names, data types and descriptive statistics. Data quality checks identify missing values, duplicate records and logical inconsistencies. Text labels are standardised, the timestamp-encoded time columns are decoded into hours, and derived features such as delay hours, cost per kilometre, distance bands and weight bands are created.

Exploratory analysis examines the distribution of delivery time and status and compares delay rate, failure rate, rating and cost across weather conditions, delivery modes, partners, vehicles, package types and regions using grouping and aggregation. A chi-square test of independence, with Cramér’s V as an effect size, measures the association between each categorical factor and

the delay flag. Pearson correlation analysis measures the relationships between distance, weight, time, cost and rating.

Visualizations in the form of histograms, bar plots, line plots, scatter plots and heatmaps are created using Matplotlib and Seaborn. Because the dataset has no calendar date, the time-series operations of the course (date ranges, resampling and interpolation) are demonstrated on a clearly labelled synthetic series in the notebook; these demonstrations are kept separate from the findings.

Table 3: Tools used

Purpose	Library / Tool
Programming language	Python
Data handling and manipulation	Pandas
Numerical operations	NumPy
Static data visualization	Matplotlib, Seaborn
Statistical analysis	SciPy
Development environment	Google Colab and Jupyter Notebook
Dataset storage and access	CSV file

Key technical choices: Pandas is used to load, inspect, clean, organise and summarise the dataset. NumPy supports numerical computation, array manipulation, indexing, slicing and universal functions, and is also used to decode the timestamp-encoded time columns. Matplotlib and Seaborn plot distributions, comparisons and relationships. SciPy provides the chi-square test. The workflow is built around the five course outcomes: NumPy fundamentals, Pandas data manipulation, data cleaning and wrangling, aggregation and time-series demonstrations, and data visualization. The analysis looks for statistical associations and descriptive patterns, not for causes of delay.

6. Data Quality Assessment and Cleaning

Data quality assessment is an essential step of exploratory data analysis because the reliability of the findings depends on the quality and consistency of the input data. Before statistical analysis and visualization, the dataset was examined for structure, missing values, duplicates and logical consistency. Two significant problems were found and repaired, and several inconsistencies were documented.

6.1 Inspection of the Dataset

The data was loaded into a Pandas DataFrame from CSV format. The first inspection covered the number of rows and columns, column names, data types and sample records. The dataset has 25,000 rows and 15 columns. Column names were stripped of whitespace, and the categorical fields (partner, package type, vehicle, mode, region, weather, status and the delay flag) were stripped, converted to lower case for checking, and then restored to readable title case, with proper names such as DHL, FedEx, EV Bike and EV Van handled explicitly.

6.2 Missing Values and Duplicates

Pandas functions were used to count null values in every column and to detect duplicate rows and duplicate column names. The dataset has no missing values, no duplicate rows and no

duplicate column labels. Because no values were missing, no imputation was required and the original values were preserved throughout.

6.3 Repairing the Time Columns

The delivery_time_hours and expected_time_hours columns were stored as text timestamps of the form 1970-01-01 00:00:00.00000000N. The recorded number of hours had been interpreted as a count of nanoseconds after the Unix epoch, so the number of hours equals the integer nanosecond value of each timestamp. The columns were decoded using NumPy's datetime64[ns] type converted to int64. After decoding, delivery time ranges from 0 to 19 hours (20 distinct values) and expected time takes nine distinct values between 2 and 24 hours. Without this repair the time columns could not be used for any numerical analysis.

6.4 The Identifier Column

The delivery_id column is stored as a decimal number ranging from 250.99 to 24,750, has only 24,502 unique values for 25,000 rows, and repeats 498 values. It is therefore not a valid unique identifier, and it carries no analytical information. It was excluded from the analysis and replaced by a sequential record number.

6.5 Consistency Checks

Logical checks were performed on the relationships between the time, delay, status, rating and cost fields. The results are summarised in Table 4.

Table 4: Data quality and consistency checks

Check	Result	Action
Missing values	0 in all 15 columns	None required
Duplicate rows	0	None required
Time columns	Stored as timestamps	Decoded to numeric hours
Identifier column	24,502 unique values, 498 repeats	Excluded; record number created
Delivered but later than expected	0 of 18,331 delivered records	Consistent
Delay flag versus status	Flag is "yes" exactly for delayed (5,341) and failed (1,328) records	Consistent
Delay flag versus times	1,203 flagged records have actual time equal to expected time	Documented; flag used as supplied
Rating versus status	Delivered 3–5, delayed 2–3, failed 1–2	Documented (Section 7.10)
Delivery cost ceiling	250 records equal the maximum 1,632.72	Documented as capped values
Outliers (IQR rule)	Delivery time: 203 records above 14 h; distance, weight and cost: none	Flagged, not removed

The 203 delivery-time outliers (more than 14 hours) occur only in adverse weather (stormy 133, rainy 60, foggy 10) on long journeys, so they are genuine extreme cases rather than errors and were retained.

6.6 Data Preparation for Analysis

After the checks, the dataset was prepared for exploratory analysis by creating derived variables: **delay_hours** (actual minus expected time), **is_delayed** and **is_failed** (0/1 indicators), **time_overrun** (actual time greater than expected time), **cost_per_km**, **distance_band** (six 50 km bands), **weight_band** (five 10 kg bands) and **rating_level**. The categorical columns were converted to the Pandas category type.

The project also demonstrates data-wrangling techniques such as replacing values, mapping, renaming, discretization, outlier detection, hierarchical indexing, merging, joining, concatenation, overlap combination and reshaping with pivot and stack operations. These operations are carried out on copies of the data so that the original dataset is preserved.

6.7 Summary

The assessment shows a complete dataset with no missing or duplicate records, but with two structural problems (timestamp-encoded times and an unusable identifier) and a few logical features that must be kept in mind when interpreting results: 1,203 delay flags that cannot be derived from the recorded times, a cost ceiling and ratings that follow the delivery status closely. The cleaned dataset has 25,000 records and 23 columns.

7. Exploratory Data Analysis

7.1 Overview of Delivery Performance

The Delivery Logistics dataset consists of 25,000 deliveries. The main performance measures are the delivery time, the delivery status and the delayed flag. The mean delivery time is 6.25 hours and the median is 6.0 hours, so the distribution is nearly symmetric with a modest right tail: 12 hours is the 95th percentile and the longest delivery took 19 hours. The expected (promised) time averages 13.11 hours, ranging from 2 to 24 hours depending on the delivery mode.

Of the 25,000 deliveries, 18,331 (73.3%) were delivered, 5,341 (21.4%) were delayed and 1,328 (5.3%) failed. The delayed flag is “yes” for the delayed and failed deliveries together, giving 6,669 flagged deliveries (26.68%). Using the recorded times alone, 5,466 deliveries (21.86%) took longer than expected, by 2.94 hours on average.

Table 5: Summary statistics of delivery time (hours)

Statistical Measure	Delivery Time (hours)
Number of deliveries	25,000
Mean	6.25
Median	6.0
Standard deviation	3.14
Minimum	0
25th percentile (Q1)	4
75th percentile (Q3)	8
90th percentile	11
95th percentile	12
99th percentile	14
Maximum	19

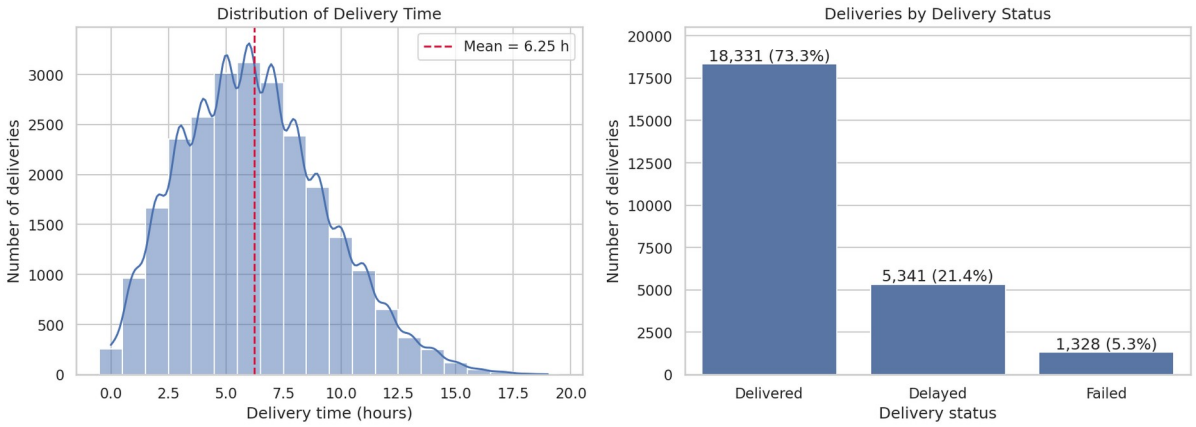


Figure 1: Distribution of delivery time (left) and number of deliveries by delivery status (right).

Table 6 summarises the other numeric variables. Distance averages 150.4 km (range 3.6–297.1 km), package weight averages 25.1 kg, delivery cost averages 864.94 and the mean customer rating is 3.67 out of 5.

Table 6: Summary of the other numeric variables

Variable	Mean	Median	Std. Dev.	Minimum	Maximum
Distance (km)	150.39	151.00	86.41	3.60	297.10
Package weight (kg)	25.15	25.15	14.37	0.67	49.52
Delivery cost	864.94	867.54	435.71	95.67	1,632.72
Customer rating	3.67	4.00	1.15	1.00	5.00

7.2 Delivery Performance by Weather Condition

Weather conditions are recorded in six categories. Table 7 compares the number of deliveries, the delivery time, the delay and failure rates and the mean rating for each condition.

Table 7: Delivery performance by weather condition

Weather	Deliveries	Mean time (h)	Median time (h)	Delay rate (%)	Failed rate (%)	Mean rating
Stormy	4,198	8.81	9.0	41.45	8.91	3.35
Rainy	4,171	7.80	8.0	37.35	7.38	3.47
Foggy	4,219	6.49	6.0	30.32	5.93	3.60
Clear	4,124	4.78	5.0	17.43	3.18	3.84
Hot	4,130	4.78	5.0	17.12	3.24	3.86
Cold	4,158	4.77	5.0	16.02	3.15	3.88

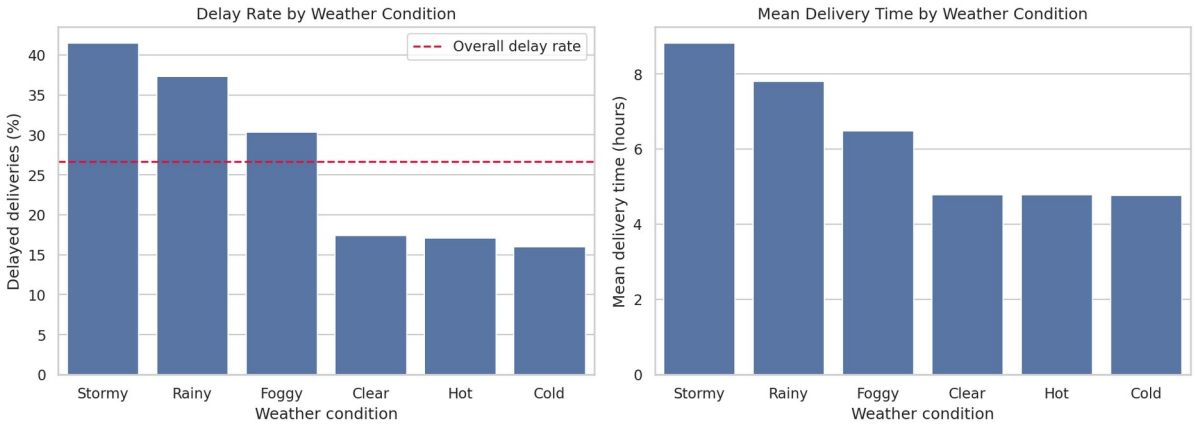


Figure 2: Delay rate (left) and mean delivery time (right) by weather condition.

Weather is clearly associated with delivery performance. Stormy weather has the highest delay rate (41.5%) and the longest mean delivery time (8.81 hours), followed by rainy (37.4%, 7.80 hours) and foggy (30.3%, 6.49 hours) conditions. Clear, hot and cold weather are similar to one another, with delay rates of 16.0–17.4% and mean times of about 4.78 hours. Failure rates and customer ratings follow the same order: 8.9% of stormy-weather deliveries failed against 3.2% in cold weather, and the mean rating falls from 3.88 to 3.35.

Two checks show that this pattern is not an artefact of other variables. The average distance is almost the same in every weather group (149.3–151.0 km), and the average expected time is also similar (12.88–13.31 hours), which means that the promised delivery window is not adjusted for weather. Delivery cost shows no relationship with weather (859.25–868.23).

7.3 Delivery Performance by Delivery Mode

Table 8: Delivery performance by delivery mode

Mode	Deliveries	Expected (h)	Actual (h)	Delay rate (%)	Failed rate (%)	Mean rating	Mean cost
Express	6,233	4.52	6.23	73.78	14.42	2.72	880.4
Same Day	6,279	8.00	6.24	32.54	6.75	3.54	929.9
Two Day	6,302	16.00	6.28	0.43	0.08	4.18	829.7
Standard	6,186	24.00	6.24	0.00	0.00	4.22	819.3

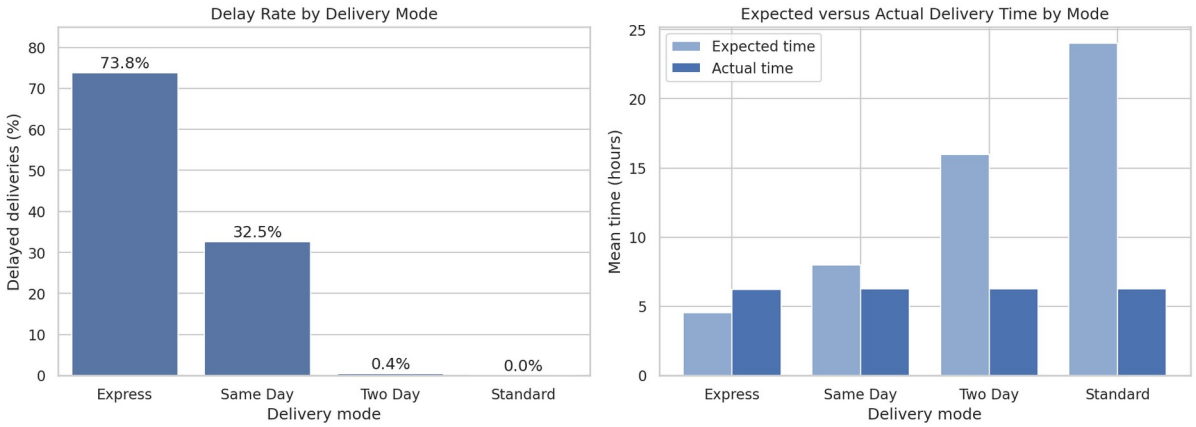


Figure 3: Delay rate by delivery mode (left) and expected versus actual mean delivery time (right).

Delivery mode shows the largest differences in the dataset. 73.8% of express deliveries and 32.5% of same-day deliveries are delayed, compared with 0.4% of two-day and 0.0% of

standard deliveries. Customer ratings follow the same order, from 2.72 for express to 4.22 for standard.

The explanation lies in the relationship between the promised and the actual time. The expected time is fixed by the mode: 8 hours for same day, 16 hours for two day and 24 hours for standard, while express promises range from 2 to 8 hours and rise with distance (from 2.01 hours for 0–50 km to 7.01 hours for 251–300 km; correlation with distance 0.99). The actual delivery time, however, is almost identical in all four modes (6.23–6.28 hours on average). A delivery therefore tends to take the same time whichever service is purchased, and the mode with the shortest promise has the highest chance of missing it. Delivery cost does not rise in step with the speed of the service, since the mean cost of an express delivery (880.4) is lower than that of a same-day delivery (929.9).

7.4 Weather and Delivery Mode Together

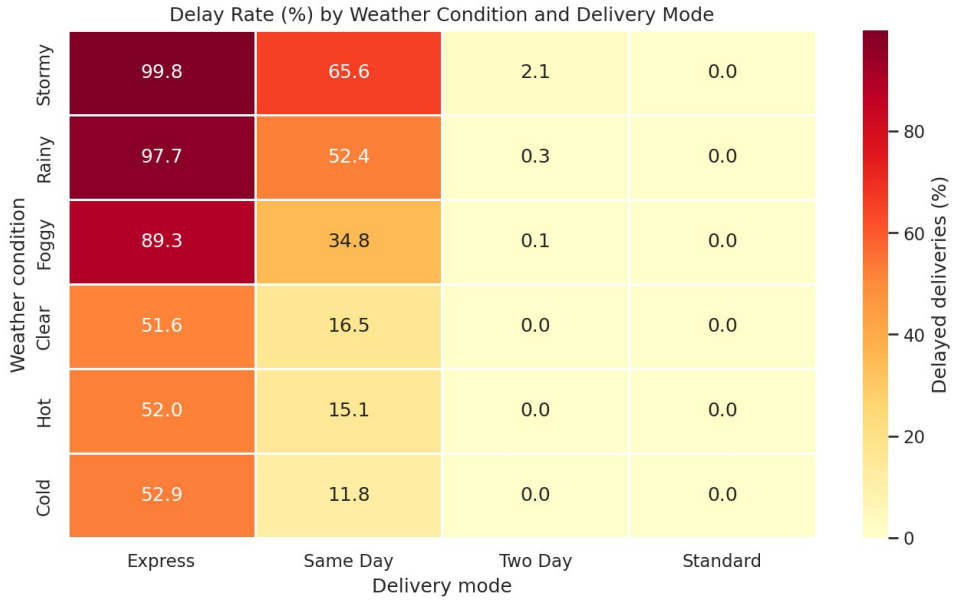


Figure 4: Delay rate (%) by weather condition and delivery mode.

Combining the two factors shows that their effects add up. Express deliveries are delayed in about half the cases even in clear, hot or cold weather (51.6%, 52.0% and 52.9%), but in 89.3% of cases in fog, 97.7% in rain and 99.8% in storms. Same-day deliveries range from 11.8% in cold weather to 65.6% in storms. Two-day deliveries are delayed in at most 2.1% of cases and standard deliveries never, because their promised windows are long compared with the actual time.

7.5 Delivery Performance by Delivery Partner

Table 9: Delivery performance by delivery partner

Partner	Deliveries	Mean time (h)	Delay rate (%)	Failed rate (%)	Mean rating	Mean cost
Xpressbees	2,826	6.32	28.27	5.91	3.61	868.2
Ekart	2,801	6.27	27.42	5.50	3.66	870.0
Shadowfax	2,736	6.27	27.30	4.86	3.66	868.3
Amazon Logistics	2,711	6.28	27.22	5.64	3.64	868.2
Blue Dart	2,798	6.30	26.95	5.79	3.68	863.0

Partner	Deliveries	Mean time (h)	Delay rate (%)	Failed rate (%)	Mean rating	Mean cost
Ecom Express	2,722	6.25	26.63	4.41	3.67	868.8
DHL	2,802	6.25	26.34	5.92	3.69	872.8
FedEx	2,818	6.14	25.16	4.76	3.70	857.4
Delhivery	2,786	6.16	24.80	4.99	3.69	848.1

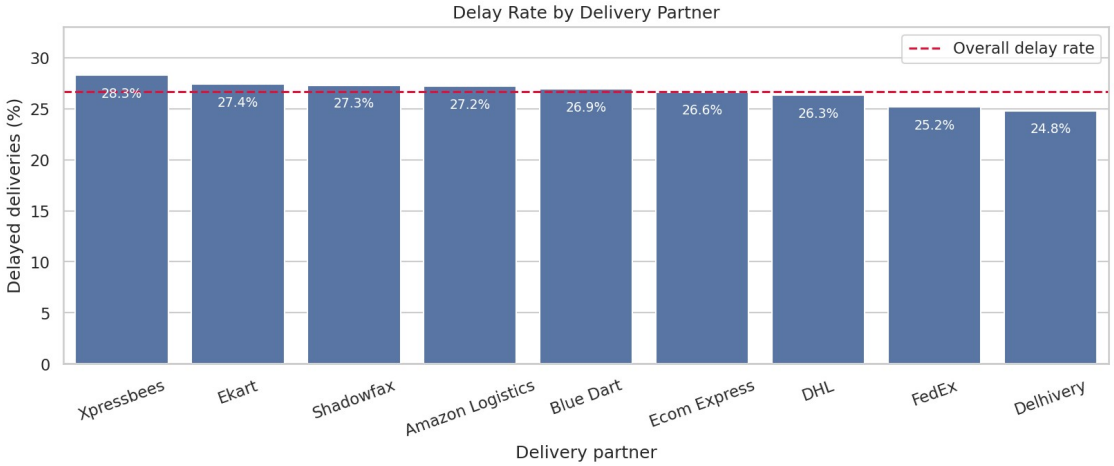


Figure 5: Delay rate by delivery partner, compared with the overall delay rate.

The nine partners handle similar numbers of deliveries (2,711–2,826), and their delay rates lie in a narrow range from 24.8% (Delhivery) to 28.3% (Xpressbees). Mean delivery time (6.14–6.32 hours), rating (3.61–3.70) and cost are also very close. A chi-square test of the partner against the delay flag gives $p = 0.081$ and Cramér's $V = 0.024$, so the partner differences are not statistically significant at the 5% level and are practically negligible.

7.6 Vehicle Type and Package Type

Table 10: Delivery performance by vehicle type

Vehicle	Deliveries	Mean time (h)	Delay rate (%)	Failed rate (%)	Mean rating	Mean cost
Bike	4,160	6.28	27.04	5.53	3.65	868.9
Truck	4,145	6.22	27.04	5.81	3.67	864.8
Van	4,187	6.27	26.77	5.40	3.66	857.7
EV Van	4,116	6.25	26.63	5.22	3.68	866.2
EV Bike	4,218	6.20	26.43	5.12	3.67	869.2
Scooter	4,174	6.28	26.14	4.79	3.68	862.9

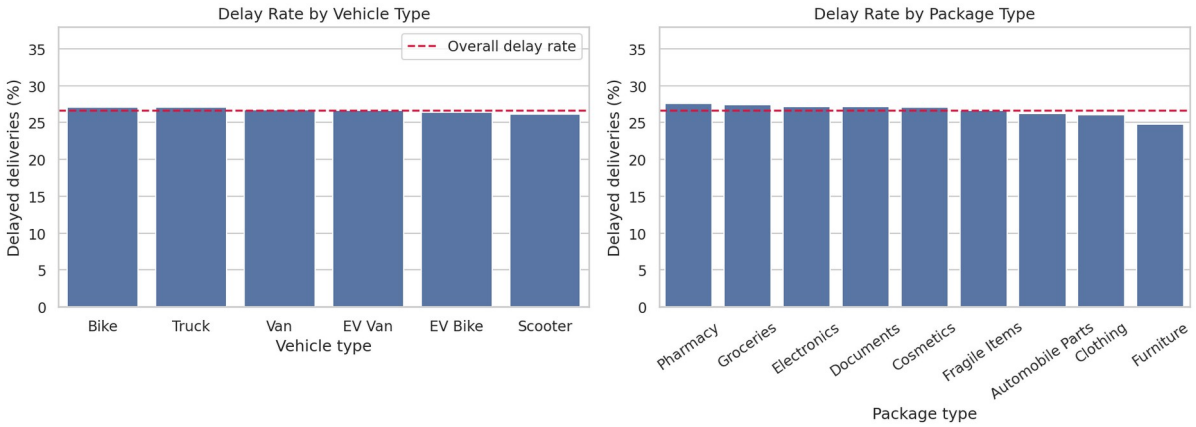


Figure 6: Delay rate by vehicle type (left) and by package type (right).

Vehicle type makes almost no difference: delay rates range only from 26.1% to 27.0%, and electric and conventional vehicles perform alike ($p = 0.930$, Cramér's $V = 0.007$). Package type behaves the same way, with delay rates from 24.8% to 27.5% ($p = 0.364$, Cramér's $V = 0.019$); the full package-type table is given in Appendix B.

7.7 Statistical Tests of Association with Delay

To check which categorical factors are genuinely associated with the delay flag, a chi-square test of independence was applied to each factor, and Cramér's V was calculated as a measure of effect size (0 = no association, 1 = perfect association).

Table 11: Chi-square tests of association with the delay flag

Factor	Chi-square	df	p-value	Cramér's V	Interpretation
Delivery mode	11,652.3	3	< 0.001	0.683	Strong association
Weather condition	1,354.5	5	< 0.001	0.233	Moderate association
Delivery partner	14.0	8	0.081	0.024	Negligible; not significant
Package type	8.7	8	0.364	0.019	Negligible; not significant
Region	3.0	4	0.550	0.011	Negligible; not significant
Vehicle type	1.3	5	0.930	0.007	Negligible; not significant

Only delivery mode and weather condition show a statistically significant and practically meaningful association with delay. The other four factors are not significant at the 5% level and have Cramér's V values close to zero. This confirms the descriptive comparisons: delay is driven by the service level promised and by the weather, not by which partner, vehicle, package type or region is involved.

7.8 Distance, Weight, Delivery Time and Cost

Table 12: Delivery performance by distance band

Distance band	Deliveries	Mean time (h)	Delay rate (%)	Failed rate (%)	Mean rating	Mean cost
0-50 km	4,101	3.10	17.02	3.34	3.86	239.6
51-100 km	4,216	4.39	19.12	4.17	3.82	488.5
101-150 km	4,102	5.62	23.94	5.34	3.72	737.5
151-200 km	4,313	6.90	28.05	5.61	3.63	988.6

Distance band	Deliveries	Mean time (h)	Delay rate (%)	Failed rate (%)	Mean rating	Mean cost
201-250 km	4,054	8.04	32.98	6.07	3.52	1,239.6
251-300 km	4,214	9.38	38.82	7.31	3.44	1,487.2

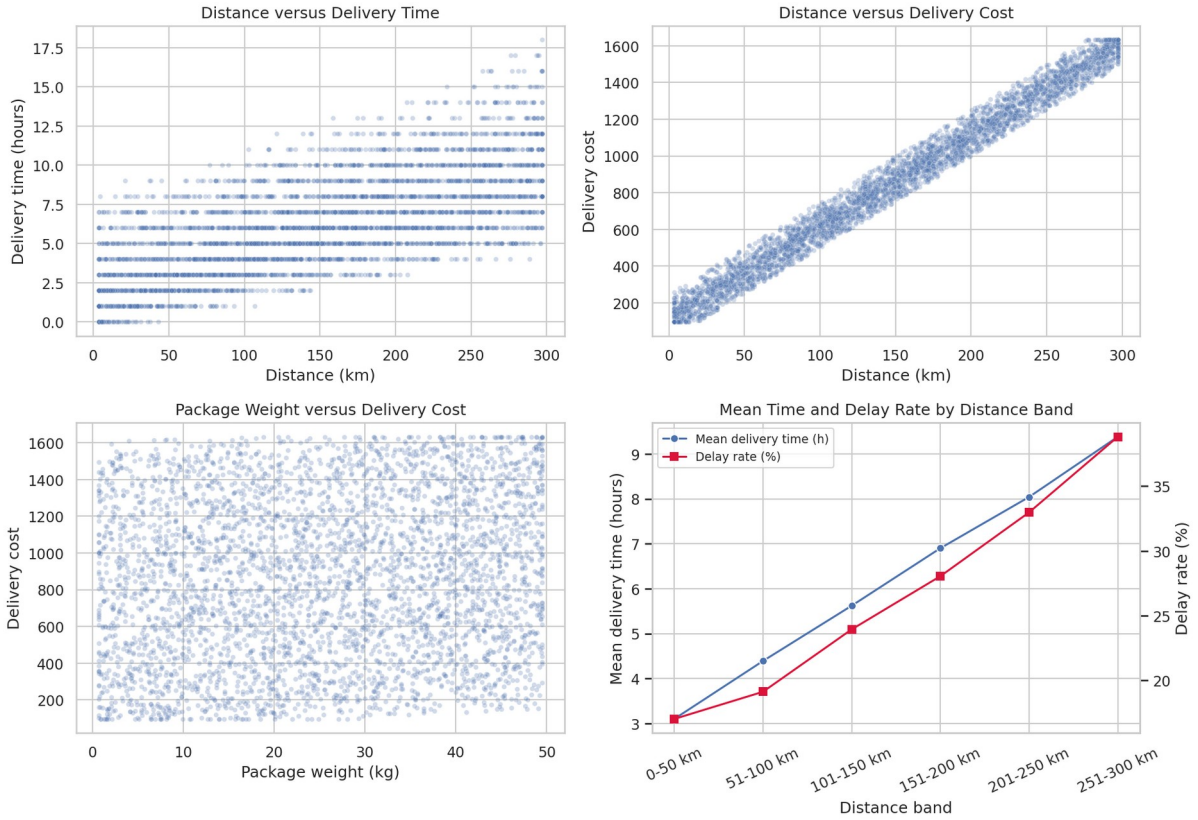


Figure 7: Relationships between distance, package weight, delivery time and delivery cost.

Distance has a clear and steady relationship with performance. Mean delivery time rises from 3.10 hours for deliveries up to 50 km to 9.38 hours for those of 251–300 km, and the delay rate rises from 17.0% to 38.8%. Delivery cost is almost a linear function of distance (r = 0.991), rising from about 240 to 1,487.

Package weight, by contrast, is unrelated to delivery time (r = 0.001) and only weakly related to cost (r = 0.099): mean cost rises from 802 for packages up to 10 kg to 925 for 41–50 kg, while mean delivery time stays near 6.2 hours in every weight band. The cost column also has a ceiling: 250 long-distance deliveries share the identical maximum cost of 1,632.72, visible as a flat upper edge in the scatter plot.

Because the promised window does not grow as quickly as the actual time, longer deliveries are late more often. For express deliveries, the promised time increases by about one hour per 50 km band, whereas the actual time (across all modes) increases by about 1.25 hours per band.

7.9 Correlation Analysis

Pearson correlation analysis was performed to examine the direction and magnitude of the linear relationships between the numeric variables. Table 13 lists the most important coefficients and Figure 8 shows the complete correlation matrix.

Table 13: Selected Pearson correlations

Variable pair	Pearson r	Interpretation
Distance – delivery cost	0.991	Very strong positive
Distance – delivery time	0.686	Strong positive
Delivery cost – delivery time	0.679	Strong positive (follows distance)
Delay flag – delivery rating	-0.777	Strong negative
Expected time – delay flag	-0.591	Moderate negative
Delivery time – delivery rating	-0.273	Weak negative
Distance – delay flag	0.174	Weak positive
Distance – delivery rating	-0.133	Very weak negative
Package weight – delivery cost	0.099	Negligible positive
Package weight – delivery time	0.001	No relationship

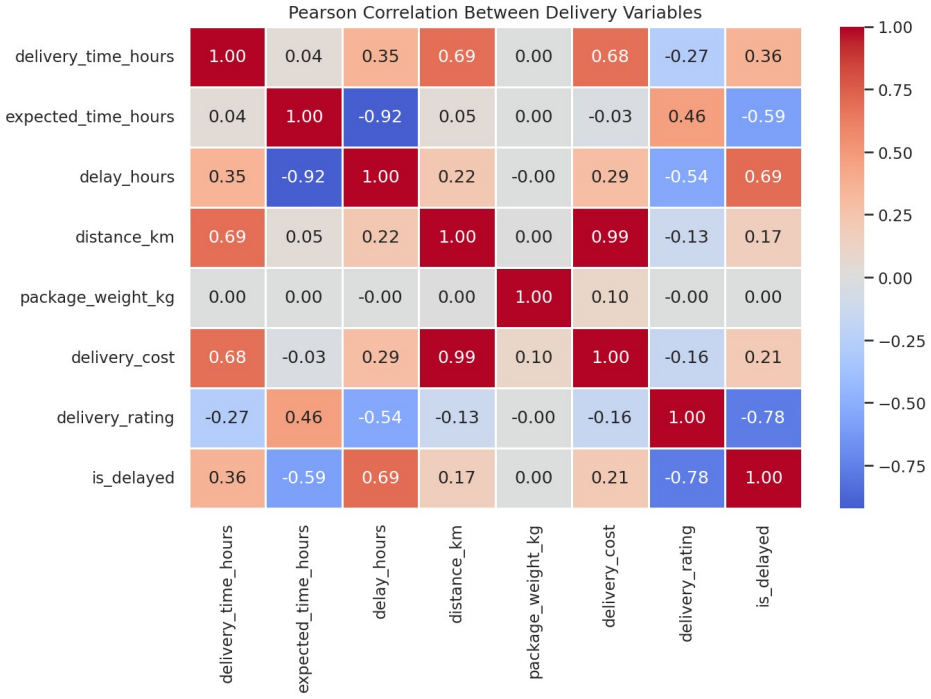


Figure 8: Pearson correlation matrix of the delivery variables.

Distance is the only variable that is strongly related to both delivery time and cost, and the strong correlation between time and cost is largely a consequence of both depending on distance. The strongest correlations involving the delay flag are with the customer rating and with the delay hours, which are themselves outcomes of the same event. Correlation does not imply causation.

7.10 Customer Rating and Delivery Status

Table 14: Delivery status, customer rating, time, distance and cost

Status	Deliveries	Rating range	Mean rating	Mean time (h)	Mean distance (km)	Mean cost
Delivered	18,331	3–5	4.21	5.57	141.3	809.8
Delayed	5,341	2–3	2.40	8.12	176.4	1,021.8

Status	Deliveries	Rating range	Mean rating	Mean time (h)	Mean distance (km)	Mean cost
Failed	1,328	1–2	1.31	8.09	171.1	995.8

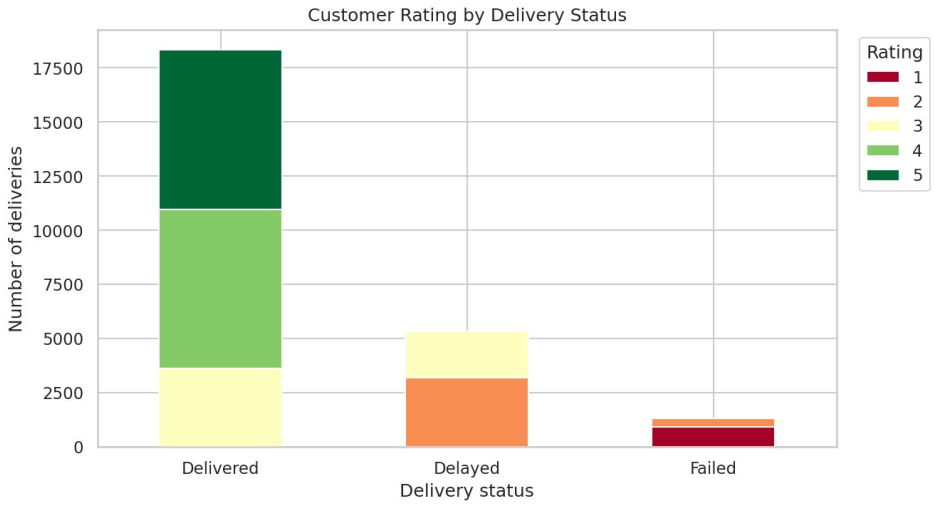


Figure 9: Customer rating by delivery status.

Ratings are closely tied to the delivery status. Delivered orders are rated 3–5 (mean 4.21), delayed orders 2–3 (mean 2.40) and failed orders 1–2 (mean 1.31), with no overlap between failed and delivered orders. The ratings therefore mirror the delivery outcome almost completely, and the delay flag and the rating should not be treated as independent pieces of evidence. The practical implication is that a delay or failure lowers the customer rating, and the delayed and failed deliveries are on average longer journeys (176 and 171 km) than delivered ones (141 km).

7.11 Summary of Exploratory Findings

- Delivery time is nearly symmetric with a mean of 6.25 hours; 26.68% of deliveries are flagged delayed and 5.31% failed.
- Adverse weather raises delay rates from about 16–17% (cold, hot, clear) to 30% (foggy), 37% (rainy) and 41% (stormy).
- Delivery mode has the strongest association with delay (express 73.8%, standard 0.0%) because actual time is the same in all modes while promised time differs.
- Delay rate rises with distance, from 17.0% to 38.8%; cost is almost linear in distance, and weight is unrelated to time.
- Partner, vehicle, package type and region show small differences that are not statistically significant.
- Customer ratings follow the delivery status closely, so delays carry a direct satisfaction cost.

8. Geospatial Visualization and Spatial Analytics

Geospatial visualization and spatial analytics help identify patterns, relationships and variations related to geographical locations. Geospatial visualization uses maps and other location-based visual elements to display information, whereas spatial analytics analyses geospatial data to understand how observations vary across space. These techniques are widely applied in transportation, logistics, public health, environmental monitoring and business intelligence.

For a courier network, spatial analysis can show where delays cluster, which corridors are slow, and how route length, terrain and local conditions affect delivery time. Maps of pick-up and drop-off points, route-level delay rates or zone-level cost would be natural outputs of such an analysis.

The present dataset does not, however, contain the data needed for a true geospatial analysis. It has no latitude or longitude, no origin or destination, no city or pin code, and no route information. The only spatial attribute is the region field, which takes five coarse values (north, south, east, west and central). Region is therefore analysed here as an ordinary categorical variable. No map is drawn, because plotting five labels on a map without any coordinates or boundaries would imply spatial detail that the data does not contain.

8.1 Regional Comparison

Table 15: Delivery performance by region

Region	Deliveries	Mean time (h)	Delay rate (%)	Failed rate (%)	Mean rating	Mean cost
Central	5,060	6.28	27.25	5.65	3.66	870.0
West	5,095	6.30	26.95	5.63	3.64	874.2
South	4,977	6.20	26.78	5.51	3.67	858.3
North	4,949	6.25	26.57	4.77	3.68	860.4
East	4,919	6.21	25.80	4.98	3.68	861.5

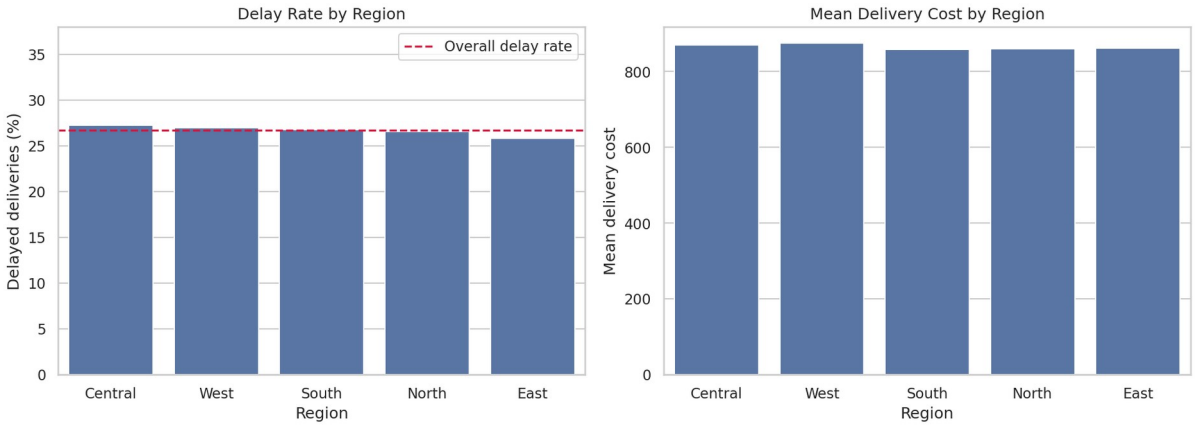


Figure 10: Delay rate (left) and mean delivery cost (right) by region.

The five regions are very similar. Delay rates lie between 25.8% and 27.3%, mean delivery times between 6.20 and 6.30 hours, and mean costs between 858 and 874. The chi-square test gives $p = 0.550$ and Cramér's $V = 0.011$, so there is no evidence of a regional effect in this dataset.

A genuine spatial analysis would require geographic attributes such as pick-up and drop-off coordinates, city or pin code, or route identifiers. Such data would allow maps of delay rate by zone and route-level analysis, and it is identified as a direction for future work (Section 13).

9. Key Findings

The exploratory analysis of the Delivery Logistics dataset revealed several important patterns in delivery performance, delay behaviour and the relationships between delivery characteristics. The major findings are summarised below.

9.1 Overall Delivery Performance

The dataset contains 25,000 deliveries with a mean delivery time of 6.25 hours and a median of 6.0 hours. 18,331 deliveries (73.3%) were delivered on time, 5,341 (21.4%) were delayed and 1,328 (5.3%) failed, so about one delivery in four is flagged as delayed.

9.2 Weather

Adverse weather is strongly associated with delay. The delay rate is 41.5% in stormy weather, 37.4% in rainy weather and 30.3% in foggy weather, compared with 16.0–17.4% in cold, hot and clear weather. Mean delivery time rises from about 4.8 hours in good weather to 8.8 hours in storms. The promised time does not change with the weather. (Cramér's V = 0.233.)

9.3 Delivery Mode

Delivery mode shows the strongest association with delay (Cramér's V = 0.683). Express deliveries are delayed 73.8% of the time and same-day deliveries 32.5%, while two-day and standard deliveries are almost never delayed. The actual time is nearly identical across modes, so the differences are determined by the length of the promised window and not by faster service.

9.4 Distance, Weight, Cost and Time

Delivery cost is almost linear in distance (r = 0.991), and delivery time rises steadily with distance (r = 0.686). The delay rate increases from 17.0% for deliveries under 50 km to 38.8% for those over 250 km. Package weight has no relationship with delivery time and only a slight relationship with cost.

9.5 Partners, Vehicles, Package Types and Regions

Differences between partners (24.8–28.3%), vehicle types (26.1–27.0%), package types (24.8–27.5%) and regions (25.8–27.3%) are small. None is statistically significant at the 5% level, and their Cramér's V values are all below 0.03.

9.6 Customer Ratings

Customer ratings follow the delivery status: delivered orders average 4.21, delayed orders 2.40 and failed orders 1.31. The correlation between the delay flag and the rating is -0.78.

9.7 Importance of Statistical Analysis and Visualization

Descriptive statistics, hypothesis tests and visualizations revealed patterns that are difficult to see in the raw table. Histograms and bar plots showed the delay and status distributions, grouped bar charts compared categories, scatter plots exposed the distance–cost relationship and the cost ceiling, a heatmap exposed the combined effect of weather and delivery mode, and the chi-square tests separated the few factors that matter from the many that do not.

9.8 Overall Interpretation

Overall, delivery performance in this dataset is shaped mainly by three things: the weather, the service level promised to the customer, and the distance travelled. Other characteristics of the delivery contribute very little. These findings describe associations within the analysed data and should not be interpreted as evidence of causal relationships.

10. Recommendations

Based on the exploratory analysis of the Delivery Logistics dataset, the following recommendations can help logistics operators better understand delivery performance and plan their operations.

1. **Make delivery promises weather-aware.** The expected time is nearly the same in every weather group, yet the delay rate rises from 16.0% in cold weather to 41.5% in stormy weather. Operators should consider extending the promised window, or adding buffer capacity, when rain, fog or storms are forecast.
2. **Review the express service level.** 73.8% of express deliveries are delayed, and about half are delayed even in clear weather, because actual delivery time is the same as for slower modes. Either the express promise should be made more realistic or the operation should give express parcels genuinely faster handling.
3. **Plan long-distance deliveries separately.** The delay rate increases from 17.0% to 38.8% as distance grows. Distance-banded promises, relay points or earlier dispatch for journeys above 200 km could reduce the gap between promised and actual time.
4. **Focus improvement effort on weather and service level, not on partner, vehicle, package type or region.** These four factors show small, statistically insignificant differences, so reallocating volume between them is unlikely to reduce delays.
5. **Communicate delays proactively.** Ratings fall sharply for delayed and failed deliveries, so early notification and recovery measures for late parcels are likely to protect customer satisfaction.
6. **Improve data capture.** Store durations as numbers, use a valid unique delivery identifier, record calendar timestamps and pick-up and drop-off locations, and avoid capping the cost field. This would allow time-series and spatial analysis and remove the repairs described in Section 6.
7. **Validate findings on operational data.** The dataset appears to be standardised, and its near-uniform category counts may not reflect a real operation. The analysis should be repeated on live operational data before decisions are taken.

11. Limitations

Although the project provides useful insights into courier delivery performance, several limitations should be considered when interpreting its results.

1. **Observational data:** The analysis identifies associations only. Associations between weather, mode or distance and delay do not prove that these factors cause delays.
2. **Data quality issues:** The time columns were stored as timestamps and the identifier column was unusable. The repairs are documented in Section 6, but the original recording errors cannot be verified.
3. **Delay flag not fully derived from times:** 1,203 deliveries are flagged as delayed or failed although their recorded actual time equals the expected time. The flag was used as supplied.
4. **Capped cost values:** 250 deliveries share the maximum cost 1,632.72, which suggests that the cost was capped. Cost relationships at long distances are therefore understated.
5. **Ratings mirror status:** Customer ratings are almost completely determined by the delivery status, so they provide little independent information.

6. **Possibly simulated or standardised data:** Category counts are nearly uniform and actual delivery time is the same for every mode. The dataset may not reflect the natural mix and behaviour of a real courier network, and the results may not generalise.
7. **Whole-hour time resolution:** Times are recorded in whole hours, which limits the precision of time comparisons.
8. **No calendar dates or coordinates:** The dataset cannot support trend, seasonality or map-based analysis. The time-series demonstrations in the notebook use synthetic data, and region is only a coarse category.
9. **Statistical significance and practical importance:** With 25,000 records even small differences can become significant. Test results were therefore interpreted together with the effect size (Cramér's V).

12. Conclusion

This project presents an exploratory analysis of courier delivery performance using the Delivery Logistics dataset, which contains 25,000 deliveries. The project applies Python-based data analysis techniques to assess and repair data quality, to examine delivery time, delay and failure rates, customer ratings and cost, to compare performance across weather conditions, delivery modes, partners, vehicles, package types and regions, and to visualize the relationships between distance, weight, time and cost.

The analysis shows a mean delivery time of 6.25 hours and a delay rate of 26.68%, with 5.31% of deliveries failing. Adverse weather raises the delay rate from 16.0% to 41.5%, and delivery mode has the strongest association with delay: 73.8% of express deliveries are delayed compared with none of the standard deliveries, because actual delivery time does not differ by mode while the promised time does.

Distance is the main driver of cost ($r = 0.991$) and an important driver of time ($r = 0.686$), whereas package weight is unrelated to time. Delivery partner, vehicle type, package type and region show small differences that are not statistically significant, and customer ratings follow the delivery status closely.

The project also demonstrates the value of exploratory data analysis and visualization in converting raw data into interpretable findings, and the importance of data quality checks: the time columns could be used only after being decoded, and the identifier column had to be set aside.

Overall, the project fulfils its objective of investigating courier delivery performance through data cleaning, descriptive statistics, statistical testing, correlation analysis and visualization. The findings should be interpreted in the context of the observational and possibly standardised nature of the data. Future work using real operational data with dates and locations, and predictive techniques, could extend this analysis.

13. Future Scope

The present project can be extended in several directions to deepen the analysis of delivery performance.

1. **Predictive modelling:** Classification models such as logistic regression, Random Forest or Gradient Boosting could predict whether a delivery will be delayed, using mode, weather, distance and expected time, and could be evaluated with suitable validation metrics.

2. **Cost modelling:** A regression model could estimate delivery cost from distance and weight and handle the capped cost values explicitly.
3. **Time-series analysis:** With reliable delivery dates and times, future work could examine daily, weekly and seasonal changes in delay rate and the effect of festivals and peak periods.
4. **Geospatial analysis:** With pick-up and drop-off coordinates, city or route data, delay hot-spots could be mapped and route-level performance compared.
5. **Service-level optimisation:** Simulation could test how weather-aware and distance-aware promised windows would change the delay rate for each delivery mode.
6. **Survival and duration analysis:** Techniques for time-to-event data could model the probability of exceeding the promised time as a function of distance and weather.
7. **Interactive dashboard:** An interactive dashboard (for example in Streamlit or Plotly) could allow operations managers to filter by weather, mode, partner and region and explore performance.
8. **Model interpretability and responsible use:** Any predictive model should include feature-importance and explanation analysis, presented carefully because feature importance does not establish causation.

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Appendix A – Notebook Structure and Implementation

The Python implementation for the Courier Delivery Performance Analysis project is provided in the Jupyter Notebook file Courier_Delivery_Performance_Analysis_Delivery_Logistics.ipynb. The notebook contains the code used for data loading, numerical computation, data inspection, cleaning, wrangling, aggregation, statistical testing and visualization. It follows the five course outcomes of the EDAV course and saves every figure used in this report to a figures folder.

The implementation is organised into the following stages:

1. **CO1 – NumPy basics:** Array creation, data types, arithmetic, indexing and slicing, Boolean and fancy indexing, transposing, random numbers and universal functions; NumPy is then applied to the real data, including decoding the timestamp-encoded hours.
2. **CO2 – Pandas and data loading:** Series, DataFrame and Index objects, reindexing, dropping, filtering, alignment, function application, sorting and ranking, duplicate checks, descriptive statistics, chunked reading and CSV writing.
3. **CO3 – Data cleaning and wrangling:** Text standardisation, decoding of time columns, handling of the identifier column, consistency checks, derived features, binning, IQR outlier flags, hierarchical indexing, joins, merges, concatenation, combine_first and pivot/stack reshaping.
4. **CO4 – Aggregation and time series:** Multiple aggregations and split-apply-combine across all factors, the weather-by-mode pivot, chi-square tests with Cramér's V, and time-series demonstrations on a labelled synthetic series.

5. **CO5 – Plotting and visualization:** Ten figures using Matplotlib and Seaborn: histograms with density curves, bar plots, grouped bars, scatter plots, a dual-axis line plot, a correlation heatmap and a weather-by-mode heatmap.
6. **Final results:** A summary generated directly from the cleaned dataset, followed by the conclusion, libraries used and run instructions.

The notebook is intended to provide a reproducible record of the analysis. The complete .ipynb file is included as a separate project deliverable.

Appendix B – Supporting Statistical Results

This appendix provides supplementary statistical information supporting the exploratory analysis of courier delivery performance.

B.1 Additional Descriptive Statistics

The distribution of delivery time is moderately right-skewed, so percentile values provide useful context alongside the mean. Table 16 gives the percentiles of delivery time that complement the summary in Table 5.

Table 16: Percentiles and extremes of delivery time (hours)

Statistical Measure	Delivery Time (hours)
25th percentile (Q1)	4
75th percentile (Q3)	8
90th percentile	11
95th percentile	12
99th percentile	14

B.2 Delivery Performance by Package Type

Table 17: Delivery performance by package type

Package type	Deliveries	Mean time (h)	Delay rate (%)	Failed rate (%)	Mean rating	Mean cost
Pharmacy	2,810	6.33	27.54	5.05	3.65	873.0
Groceries	2,693	6.23	27.44	5.24	3.66	861.9
Electronics	2,792	6.19	27.18	5.01	3.65	857.7
Documents	2,805	6.16	27.17	5.38	3.67	850.8
Cosmetics	2,744	6.29	27.04	5.39	3.64	870.8
Fragile Items	2,848	6.15	26.65	5.23	3.66	847.5
Automobile Parts	2,795	6.26	26.19	5.76	3.69	877.2
Clothing	2,767	6.40	26.09	5.49	3.69	879.4
Furniture	2,746	6.22	24.76	5.24	3.69	866.7

B.3 IQR Outlier Analysis

Table 18: Outlier analysis using the IQR rule

Variable	Q1	Q3	Lower bound	Upper bound	Outliers
delivery_time_hours	4.00	8.00	-2.00	14.00	203
distance_km	75.90	224.90	-147.60	448.40	0
package_weight_kg	12.68	37.66	-24.79	75.13	0
delivery_cost	490.80	1,237.91	-629.87	2,358.58	0

The coefficients and test results in this report show that delivery mode and weather are the only categorical factors with a meaningful association with delay, and that distance is the only numeric variable strongly related to both time and cost. Correlation does not establish causation.

Appendix C – Instructions to Run the Project

C.1 Requirements

The project uses Python and the following libraries:

- NumPy for numerical computation.
- Pandas for data manipulation and analysis.
- Matplotlib for plotting and visualization.
- Seaborn for statistical graphics.
- SciPy for the chi-square tests.

The project files are:

1. Courier_Delivery_Performance_Analysis_Delivery_Logistics.ipynb – Jupyter Notebook containing the analysis code.
2. Delivery_Logistics.csv – Dataset used for the analysis.

C.2 Running the Notebook in Google Colab

1. Open Google Colab and upload the notebook file.
2. Upload Delivery_Logistics.csv using the Files panel (the notebook looks for it in the working folder and in /content/).
3. Confirm that the dataset is available at one of the file paths expected by the notebook.
4. Run the notebook cells sequentially from top to bottom.
5. Inspect the displayed statistics, tables and visualizations; the figures are saved in the figures folder.
6. If the runtime is restarted, upload the dataset again if it is no longer available.
7. Save or download the notebook after execution if the output cells need to be included in the submitted copy.

C.3 Expected Results

Successful execution should display the dataset dimensions (25,000 rows and 15 columns), descriptive statistics, data quality checks, category-wise performance summaries, chi-square

test results, ten exploratory visualizations and the correlation matrix. The analysis uses the delivery time, the delay flag and the delivery status as the principal measures of performance.

C.4 Reproducibility Notes

The analysis depends on the availability of the original dataset and the required Python libraries. Results may vary if the dataset is changed or if the code is modified. Random sampling used for the scatter plots uses a fixed random seed (42) to improve reproducibility, and the full dataset is used for all statistics. The notebook and dataset should be retained together so that the analysis can be rerun without reconstructing the input data.